Ryerson University

Department of Electrical, Computer, & Biomedical Engineering
Faculty of Engineering & Architectural Science

Course Title	**ELE827**
Course Name	**CMOS Mixed Signal**
Semester/Year	**F2026**

Instructor	**Dr Fei Yuan**

Lab Number	**Lab 4**
Lab Name	**SAR**

Submission Date	
Due Date	

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*By signing above you attest that you have contributed to this written lab report and confirm that all work you have contributed to this lab reports your own work. Any suspicion of copying or plagiarism in this work will result in an investigation of Academic Misconduct and may result in a "0" on the work, an "F" in the course, or possibly more severe penalties, as well as a Disciplinary Notice on your academic record under the Student Code of Academic Conduct, which can be found online at: http://www.ryerson.ca/senate/current/pol60.pdf

Table of Contents:

Component Values - P3
Schematic - P3
SAR Testbench - P4
Conclusion - P6

List of Figures

Figure 1: SAR Full Schematic
Figure 2: SAR Schematic of one segment
Figure 3: SAR Testbench
Figure 4: Reset Signal
Figure 5: Clock Edges
Figure 6: Dn+ with COMP+ = 0V, COMP- = 1V
Figure 7: Dn- with COMP+ = 0V, COMP- = 1V
Figure 8: Dn+ with COMP+ = 1V, COMP- = 0V
Figure 9: Dn- with COMP+ = 1V, COMP- = 0V

List of Tables

Table 1: Component Values

List of Abbreviations

Dn+: Positive Tuning Bits

Dn-: Negative Tuning Bits

SH: The Sample and Hold Clock Signal

COMP+/COMP-: Comparator Output

C1…C6: Clock Edges Generated by Delay Locked Loop

SAR: Successive Approximation Register

Component Values

Component	PMOS Sizing	NMOS Sizing
DFF (Transmission Gate)	320n	480n
DFF (NOR2)	960n	320n
INVERTER	960n	320n
NAND2	720n	320n

Table 1: Component Values

Schematics

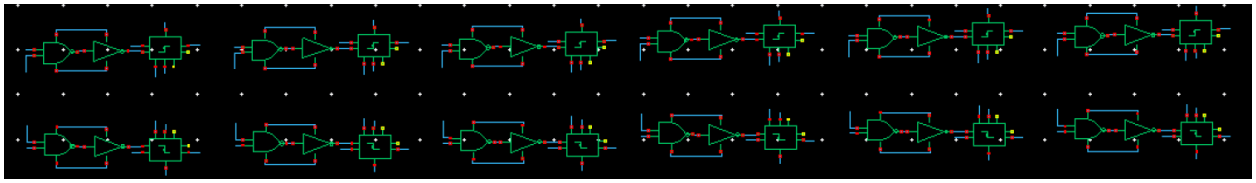


Figure 1: SAR Full Schematic

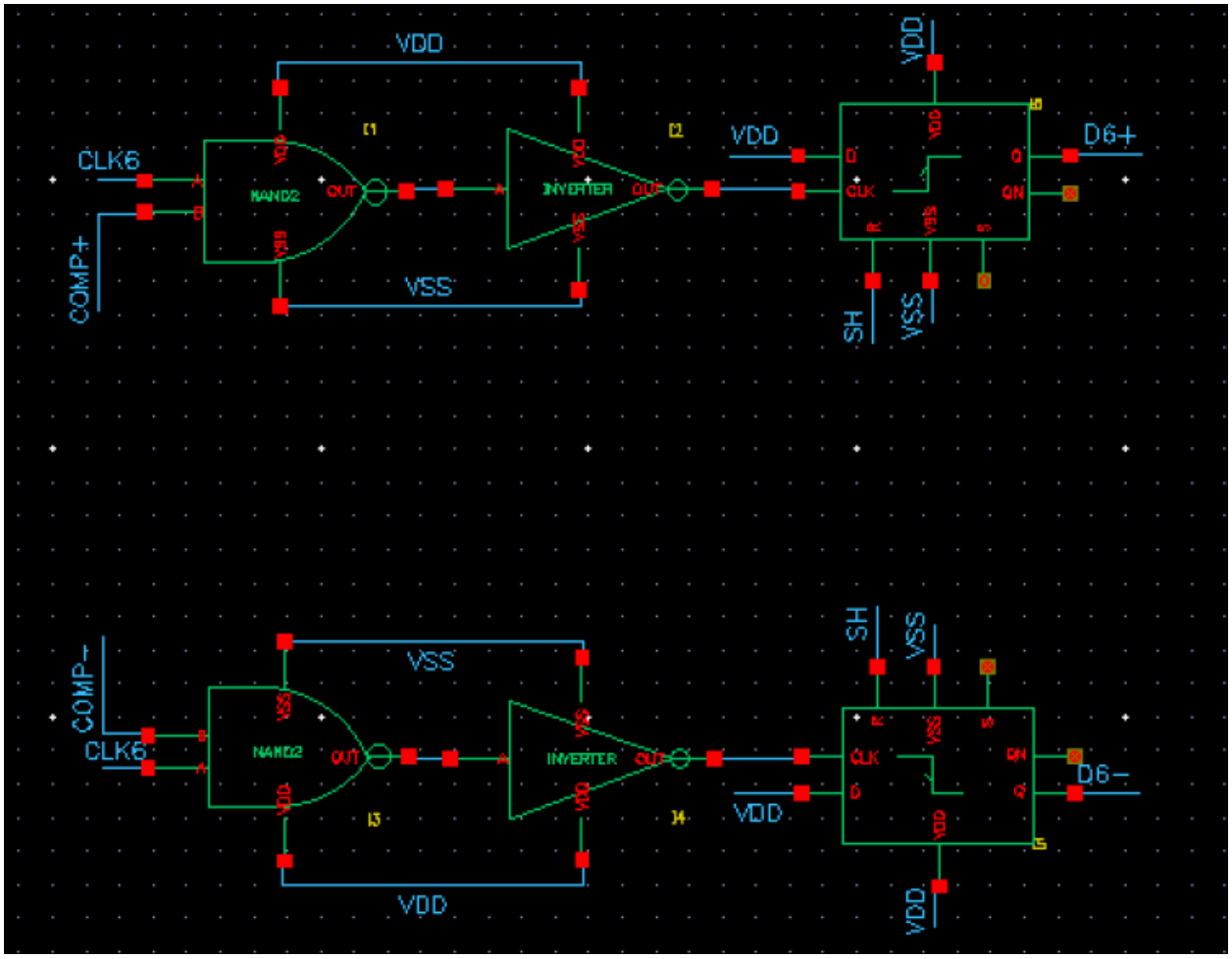


Figure 2: SAR Schematic of one segment

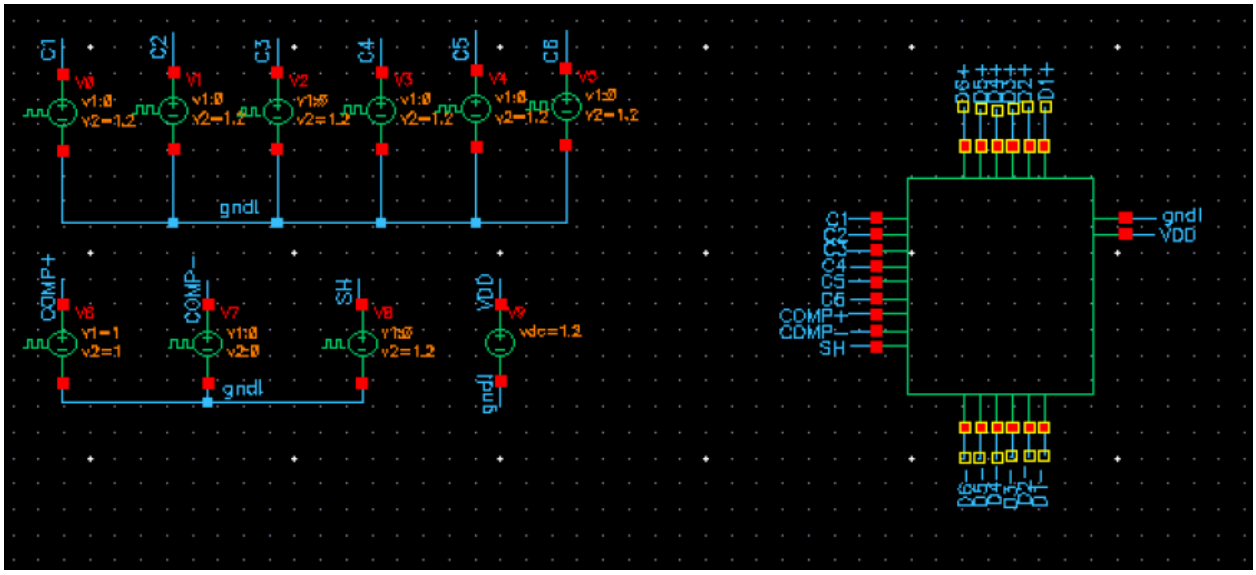


Figure 3: SAR Testbench

SAR Testbench

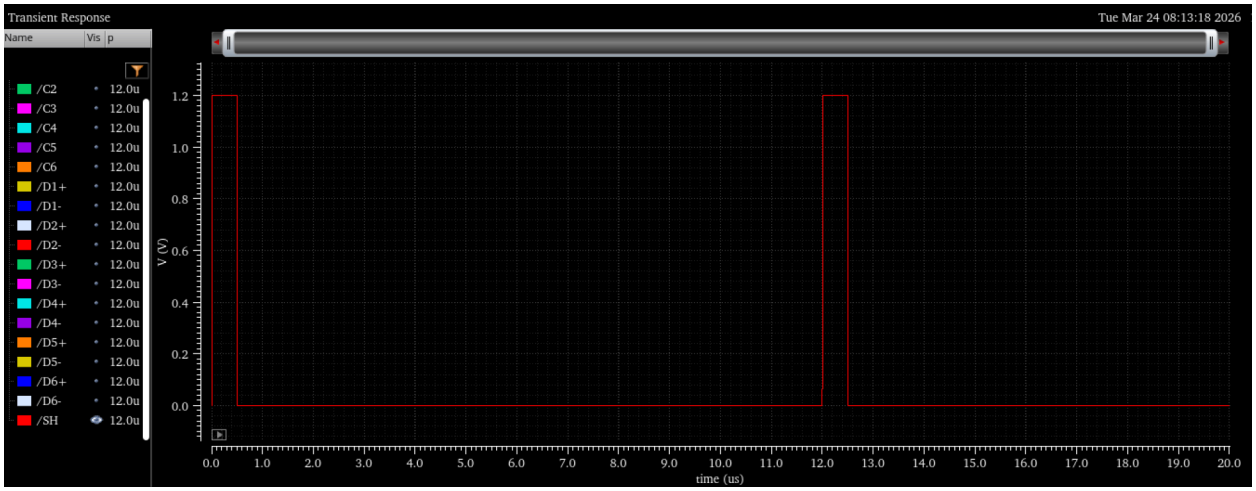


Figure 4: Reset Signal

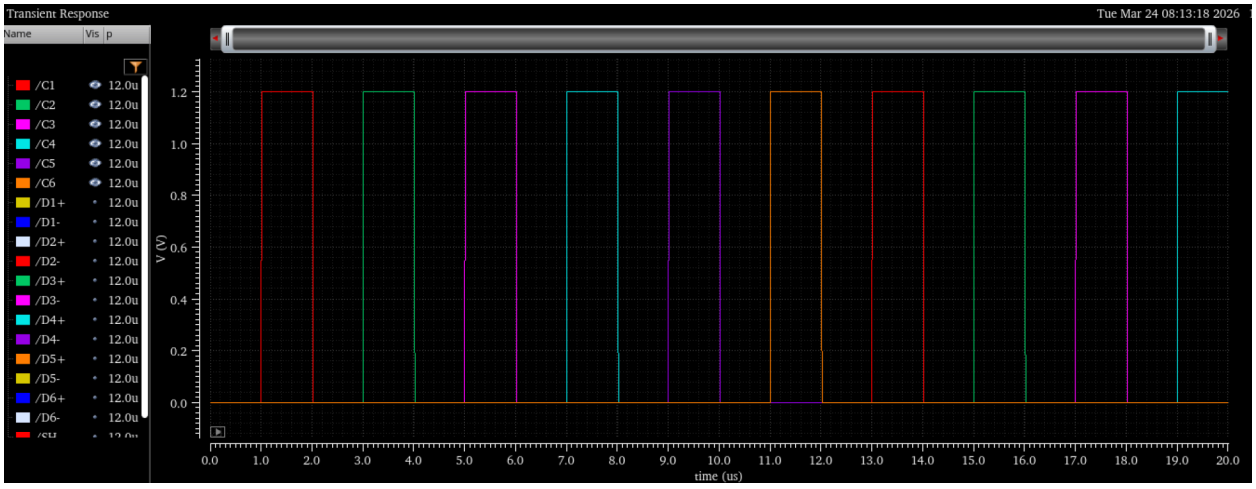


Figure 5: Clock Edges

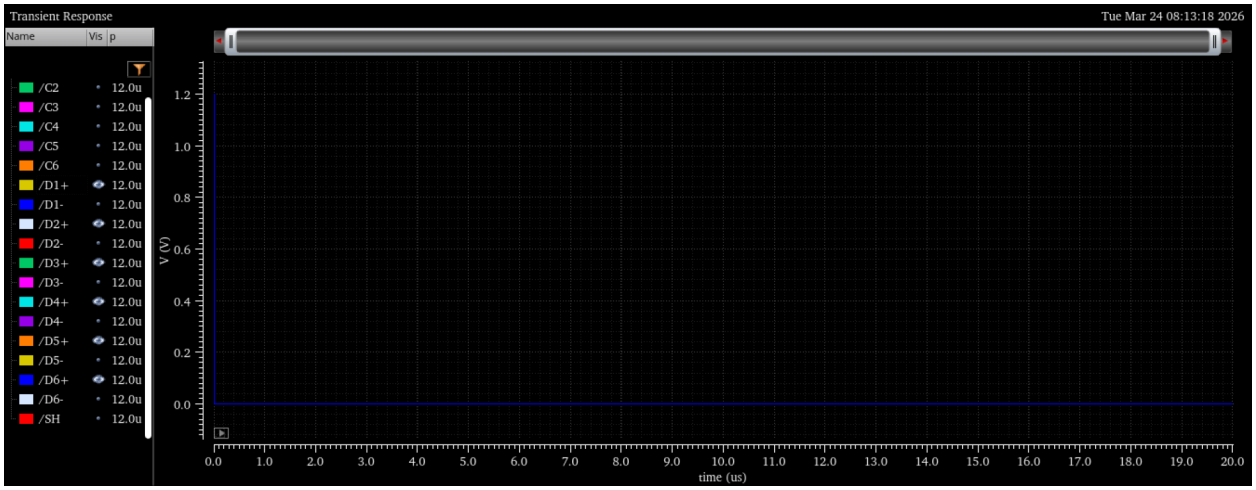


Figure 6: Dn+ with COMP+ = 0V, COMP- = 1V

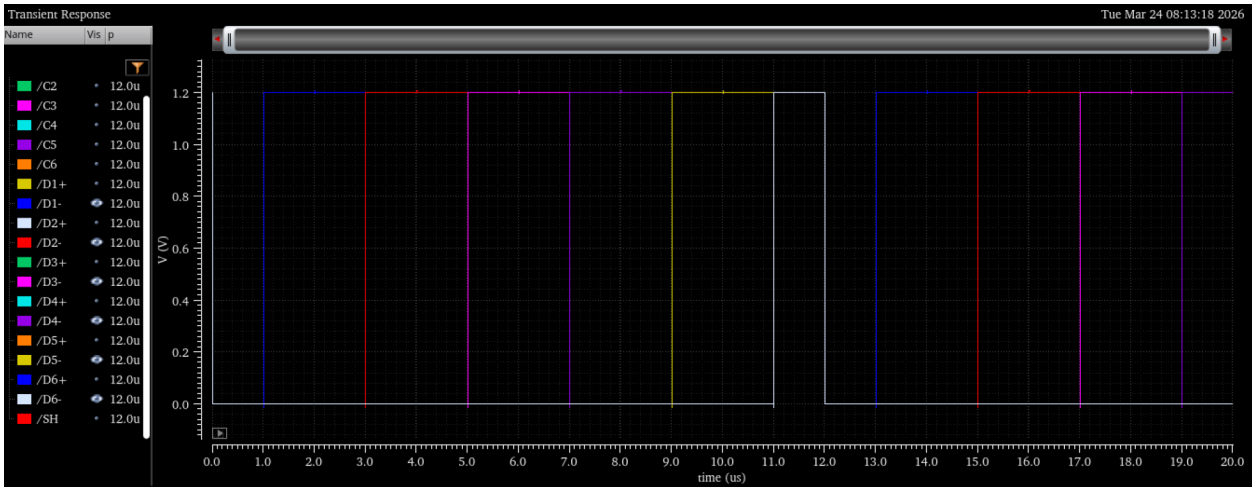


Figure 7: Dn- with COMP+ = 0V, COMP- = 1V

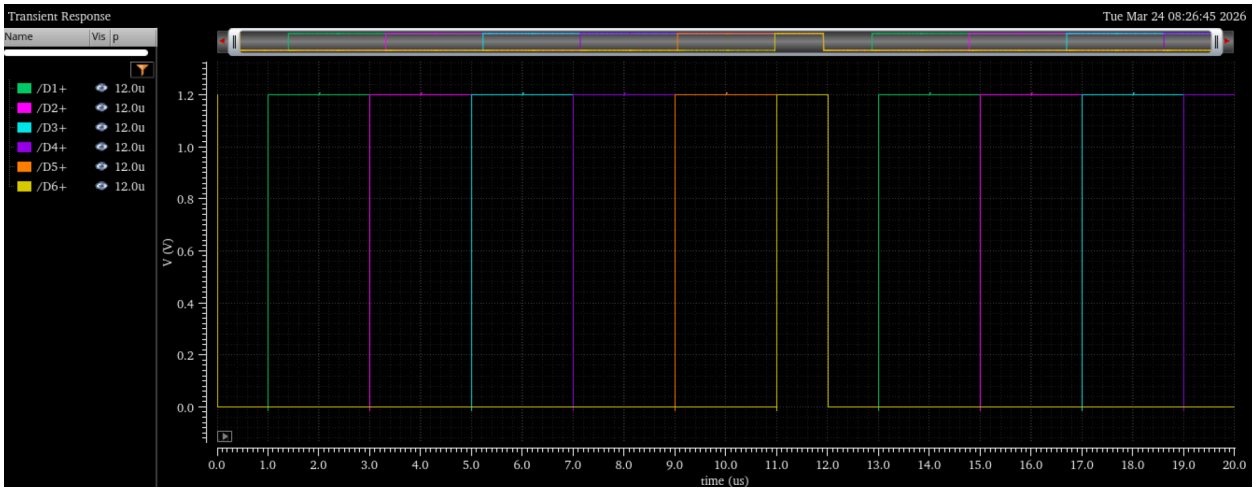


Figure 8: Dn+ with COMP+ = 1V, COMP- = 0V



Figure 9: Dn- with COMP+ = 1V, COMP- = 0V

Conclusion

In conclusion, this lab explored the behaviors of the SAR component in an SAR ADC. The SAR is tested with the comparator positive output held at 1V and the negative output held at 0V, it was observed that the positive tuning bit would slowly raise from 000000 to 100000 then to 111111 after all 6 CLK edges while the negative tuning bits would stay at 000000. When the positive comparator output is held at 0V and the negative at 1V, it can be seen the opposite happens when the negative tuning would slowly raise from 000000 to 100000 then to 111111 after all 6 CLK edges while the positive tuning bits would stay at 000000. This behavior would change if the comparator output changes between each of the CLK edges, allowing the tuning bits to go up and down depending on the output.